

Low Consumption Current High PSRR

300mA CMOS Voltage Regulator

CE6206 Series

■ INTRODUCTION

The CE6206 series are a group of positive voltage regulators manufactured by CMOS technologies with high ripple rejection, low power consumption and low dropout voltage, which can prolong battery life in portable electronics. The CE6206 series work with low-ESR ceramic capacitors, reducing the amount of board space necessary for power applications. The CE6206 series consume less than 0.1uA in shutdown mode and have fast turn-on time less than 50us. The series are very suitable for the battery-powered equipments, such as RF applications and other systems requiring a quiet voltage.

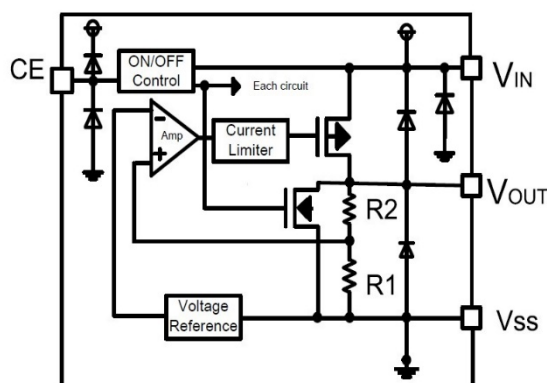
■ APPLICATIONS

- Cellular and Smart Phones
- Laptop, Palmtops and PDA
- Digital Still and Video Cameras

■ FEATURES

- Low Dropout Voltage:
150mV @ 150mA
- Low Quiescent Current: 5μA
- High Ripple Rejection: 70dB @ 1kHz
- Excellent Line and Load Transient Response
- Operating Voltage: 2.0V ~ 7.0V
- Output Voltage: 1.2V ~ 5.0V
- High Accuracy:
±2% (Typ.)/±1% (Typ.)
- Built-in Current Limiter, Short-Circuit Protection
- TTL- Logic-Controlled Shutdown Input
- MP3, MP4 Player
- Radio control systems
- Battery-Powered Equipments

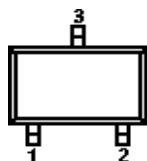
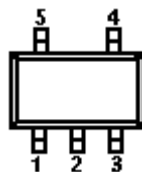
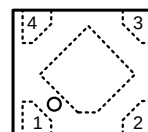
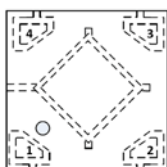
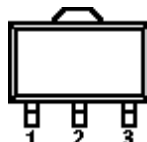
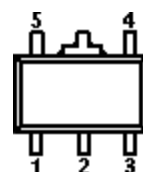
■ BLOCK DIAGRAM



■ ORDER INFORMATION

DESIGNATOR	SYMBOL	DESCRIPTION
CE6206①②③④⑤	①	A Standard
		B With Shutdown Function
	②	Integer
	③	
	④	M Package: SOT-23-3/5
		U Package: SC-70-5
		P/PT Package: SOT-89-3/5
		T Package: TO-92
		F/ FB Package: DFN1X1-4/FBP1X1-4
	⑤	- 2% Accuracy
		1 1% Accuracy

PIN CONFIGURATION (Top view)

SOT-23-3
**SOT-23-5/
SOT-353(SC70-5)**
**DFN1X1-4****FBP1X1-4****SOT-89-3****SOT-89-5**

PIN NUMBER							PIN NAME	FUNCTION
SOT-23-3				SOT-89-3		TO-92		
M	MA	MC	MY	P	PT	T		
1	2	3	3	1	2	1	V _{SS}	Ground
2	1	2	1	3	1	3	V _{OUT}	Output
3	3	1	2	2	3	2	V _{IN}	Power input

SOT-23-5/SC-70-5

PIN NUMBER	SYMBOL	FUNCTION
1	V _{IN}	Power Input Pin
2	V _{SS}	Ground
3	CE	Chip Enable Pin
4	NC	No Connection
5	V _{OUT}	Output Pin

DFN1X1-4(CE6206BXXF)/FBP1X1-4(CE6206BXXFB)

PIN NUMBER	SYMBOL	FUNCTION
F		
1	V _{OUT}	Output Pin
2	V _{SS}	Ground
3	CE	Chip Enable Pin
4	V _{IN}	Power Input Pin
EP	Thermal PAD	Ground

SOT-89-5

PIN NUMBER		SYMBOL	FUNCTION
P	PT		
1	5	V_{OUT}	Output Pin
2	2	V_{SS}	Ground
3	1	NC	No Connection
4	3	CE	Chip Enable Pin
5	4	V_{IN}	Power Input Pin

■ ABSOLUTE MAXIMUM RATINGS

(Unless otherwise specified, $T_A = 25^{\circ}\text{C}$)

PARAMETER		SYMBOL	RATINGS	UNITS
Input Voltage		V_{IN}	$V_{SS} - 0.3 \sim V_{SS} + 8$	V
Output Current		I_{OUT}	600	mA
Output Voltage		V_{OUT}	$V_{SS} - 0.3 \sim V_{IN} + 0.3$	V
Power Dissipation	SC-70-5	P_d	250	mW
	SOT-23-3/5		400	
	DFN1X1-4/FBP1X1-4		400	
	TO-92		500	
	SOT-89-3/5		600	
Operating free air temperature range		T_A	-40 ~ +85	$^{\circ}\text{C}$
Operating Junction Temperature Range		T_j	-40 ~ +150	
Storage Temperature		T_{stg}	-55 ~ +150	
Lead Temperature(Soldering, 10sec)		T_{solder}	260	
ESD rating: Human Body Model		HBM	≥ 2	kV
ESD rating: Machine Model		MM	≥ 200	V

■ ELECTRICAL CHARACTERISTICS

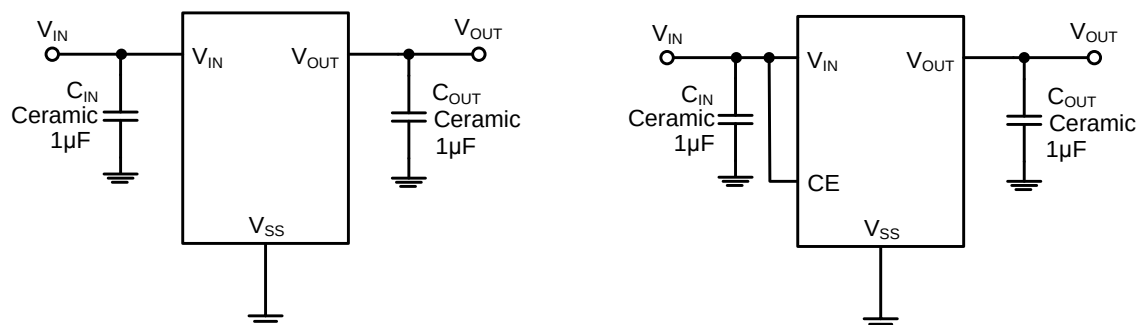
CE6206 Series ($V_{IN} = V_{OUT} + 1V$, $C_{IN} = C_{OUT} = 1\mu F$, $T_A = 25^\circ C$, unless otherwise specified)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
Output Voltage	$V_{OUT(E)}^{(2)}$	$I_{OUT} = 1mA$	$V_{OUT} * 0.98$	$V_{OUT}^{(1)}$	$V_{OUT} * 1.02$	V
			$V_{OUT} * 0.99$	V_{OUT}	$V_{OUT} * 1.01$	V
Supply Current	I_{SS}	$I_{OUT} = 0$		5	10	μA
Standby Current	I_{STBY}	$CE = V_{SS}$			0.2	μA
Output Current	I_{OUT}	—	300			mA
Dropout Voltage ⁽³⁾	V_{dif}	$I_{OUT} = 150mA$ $V_{OUT} \geq 3.0V$		150		mV
Load Regulation	ΔV_{OUT}	$V_{IN} = V_{OUT} + 1V$, $1mA \leq I_{OUT} \leq 100mA$		10		mV
Line Regulation	$\frac{\Delta V_{OUT}}{V_{OUT} \times \Delta V_{IN}}$	$I_{OUT} = 10mA$ $V_{OUT} + 1V \leq V_{IN} \leq 6V$		0.01	0.2	%/V
Output Voltage Temperature Characteristics	$\frac{\Delta V_{OUT}}{\Delta T \times V_{OUT}}$	$I_{OUT} = 10mA$ $-40^\circ C \leq T \leq +85^\circ C$		100		ppm/ $^\circ C$
Short Current	I_{Short}	$V_{OUT} = V_{SS}$		50		mA
Input Voltage	V_{IN}	—	2.0		7.0	V
Power Supply Rejection Rate	1kHz	PSRR	$I_{OUT} = 50mA$	70		dB
	10kHz			50		
CE "High" Voltage	$V_{CE"H"}$		1.5		V_{IN}	V
CE "Low" Voltage	$V_{CE"L"}$				0.3	V

NOTE:

1. V_{OUT} : Specified Output Voltage.
2. $V_{OUT(E)}$: Effective Output Voltage (I.e.The Output Voltage When $V_{IN} = (V_{OUT} + 1.0V)$, And Maintain A Certain I_{OUT} Value).
3. V_{dif} : The Difference Of Output Voltage And Input Voltage When Input Voltage Is Decreased Gradually Till Output Voltage Equals To 98% Of $V_{OUT(E)}$.

■ TYPICAL APPLICATION



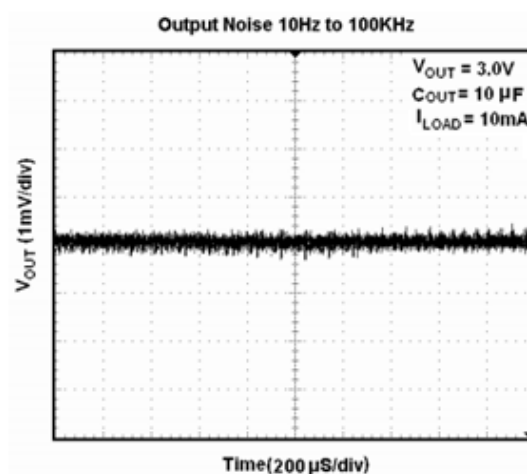
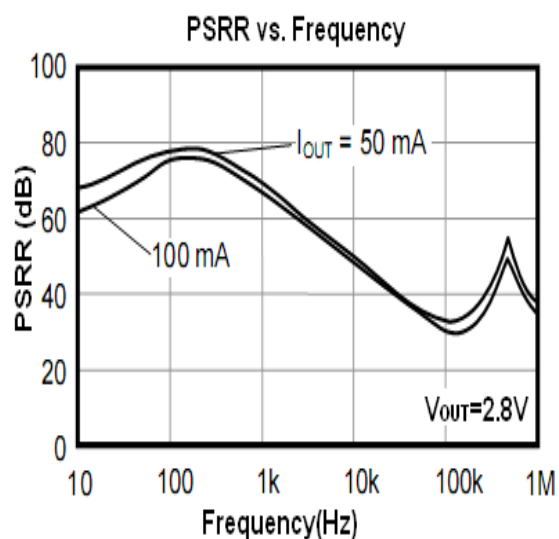
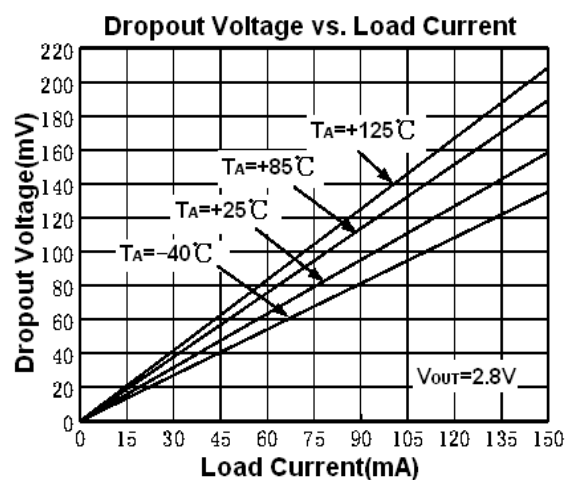
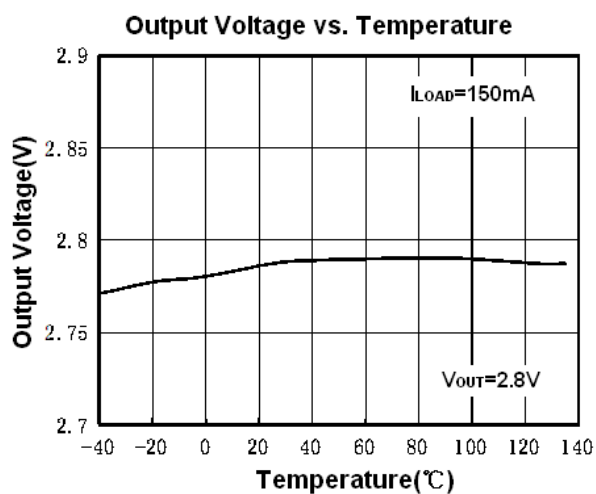
NOTE:

C_{IN} : 1.0 μF or more

C_{OUT} : 1.0 μF or more

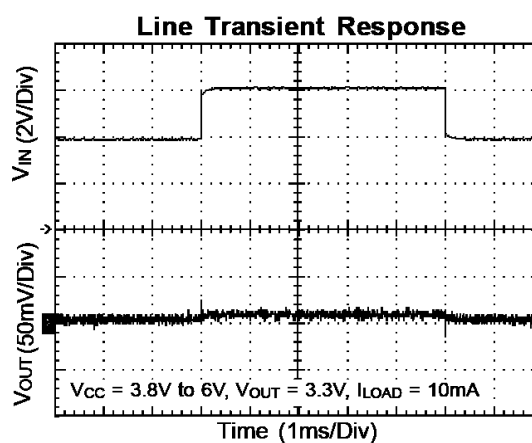
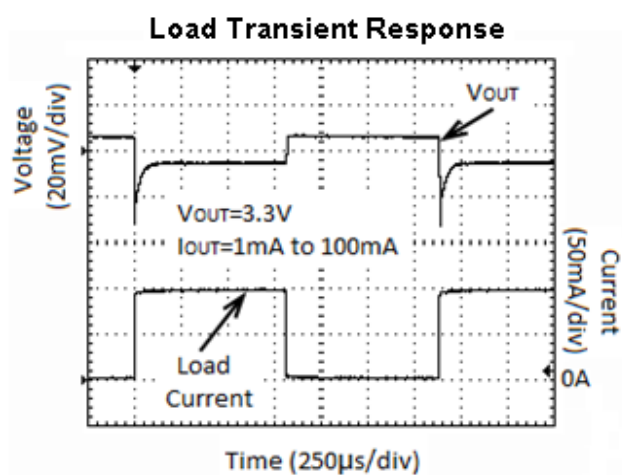
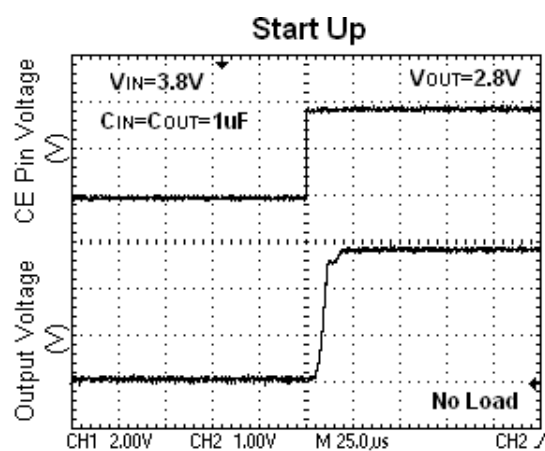
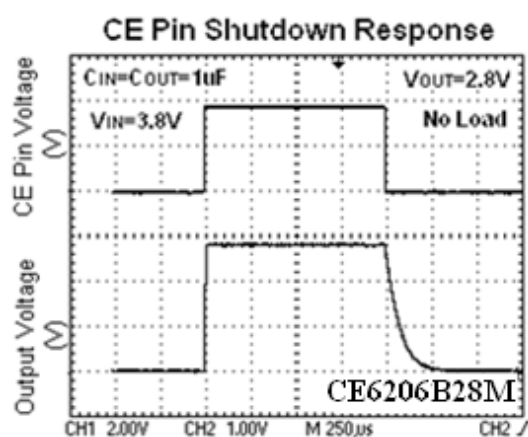
■ TYPICAL PERFORMANCE CHARACTERISTICS

($V_{CE} = V_{IN} = V_{OUT} + 1V$, $C_{IN} = C_{OUT} = 1\mu F$, $T_A = 25^\circ C$, unless otherwise specified)

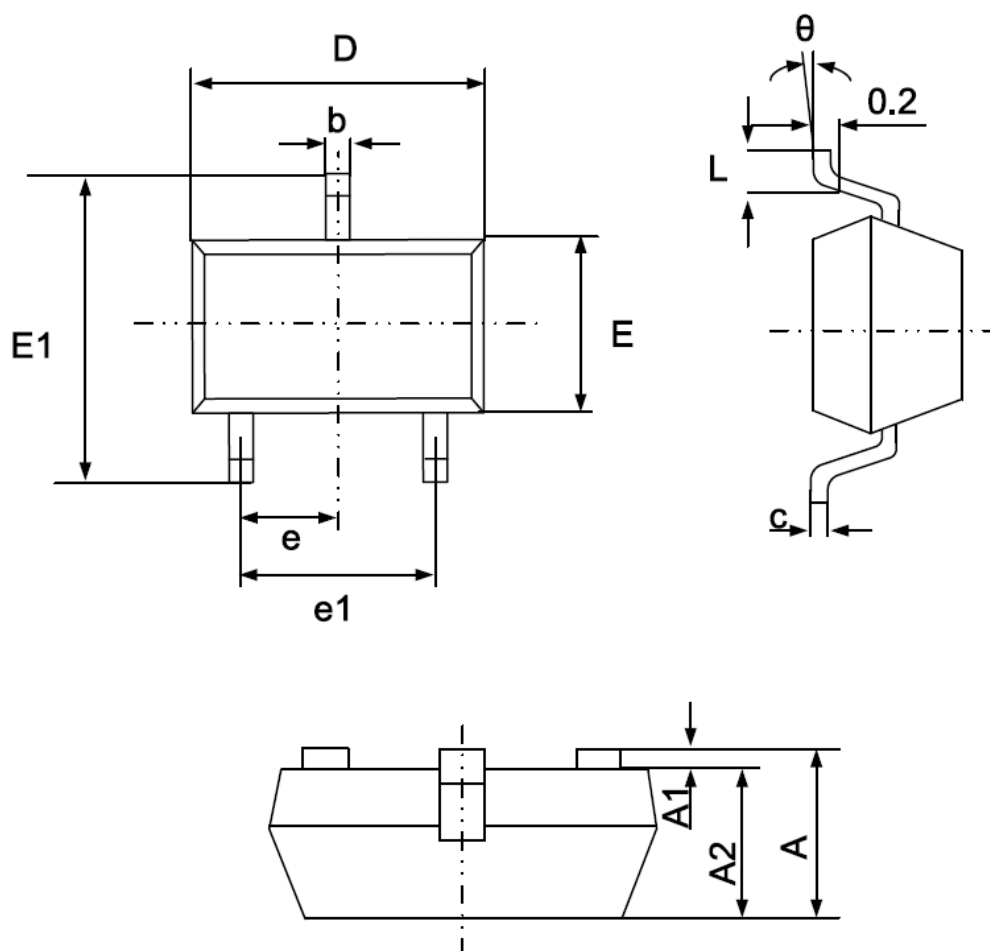


■ TYPICAL PERFORMANCE CHARACTERISTICS

($V_{CE} = V_{IN} = V_{OUT} + 1V$, $C_{IN} = C_{OUT} = 1\mu F$, $T_A = 25^\circ C$, unless otherwise specified)

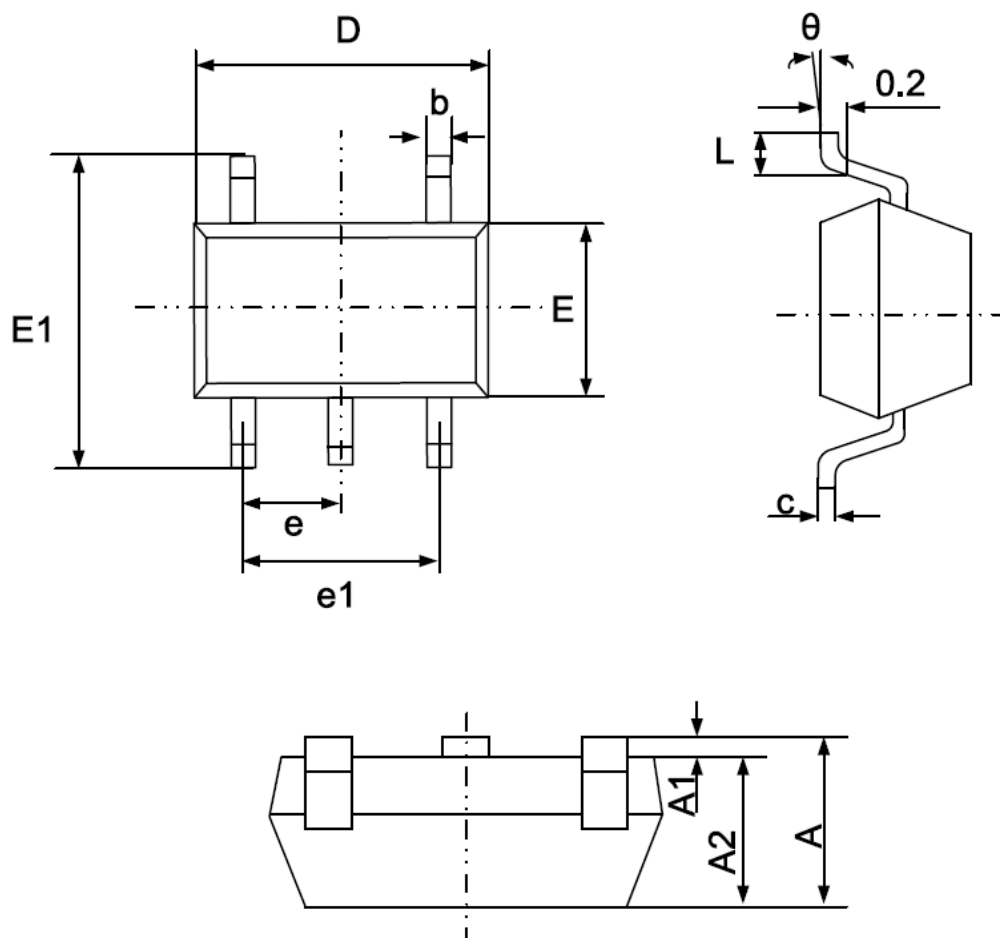


- PACKAGING INFORMATION
- SOT-23-3 PACKAGE OUTLINE DIMENSIONS



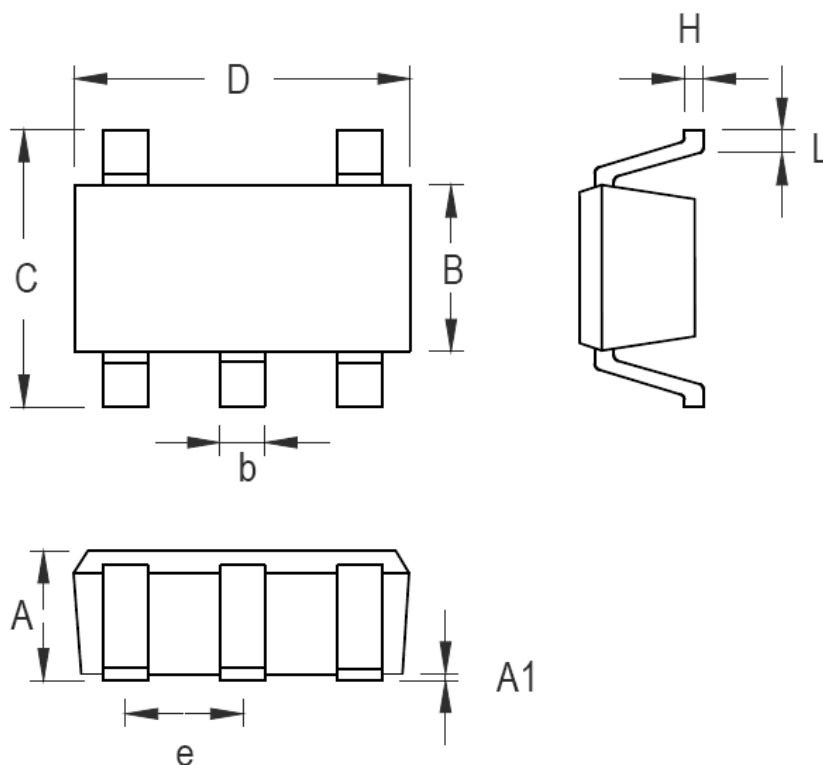
Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	1.050	1.250	0.041	0.049
A1	0.000	0.100	0.000	0.004
A2	1.050	1.150	0.041	0.045
b	0.300	0.500	0.012	0.020
c	0.100	0.200	0.004	0.008
D	2.820	3.020	0.111	0.119
E	1.500	1.700	0.059	0.067
E1	2.650	2.950	0.104	0.116
e	0.950(BSC)		0.037(BSC)	
e1	1.800	2.000	0.071	0.079
L	0.300	0.600	0.012	0.024
θ	0°	8°	0°	8°

● SOT-23-5 PACKAGE OUTLINE DIMENSIONS



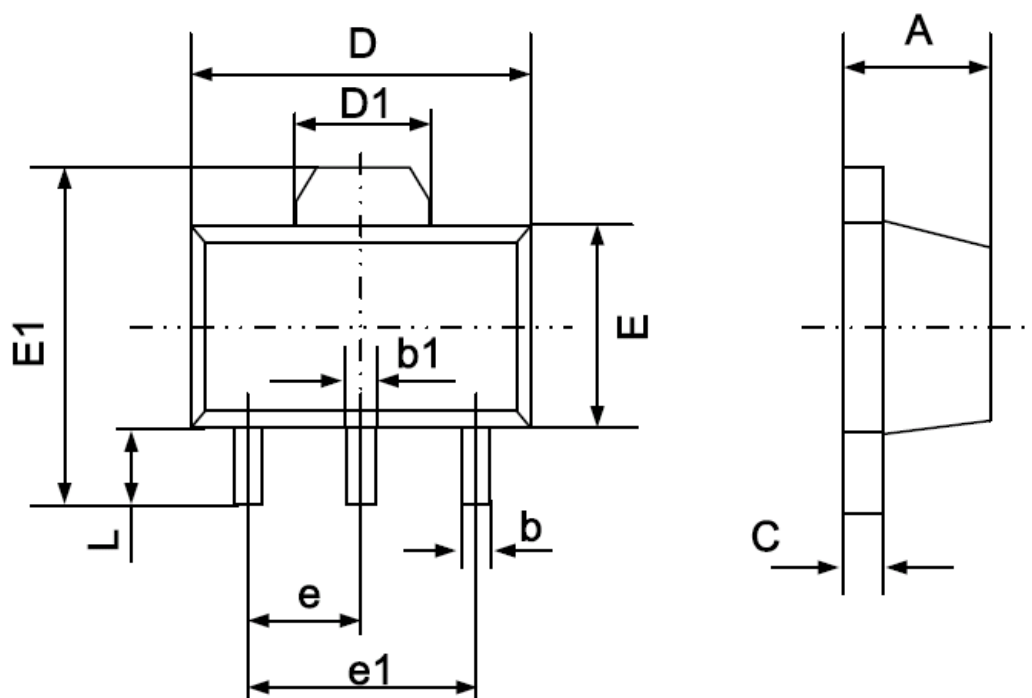
Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	1.050	1.250	0.041	0.049
A1	0.000	0.100	0.000	0.004
A2	1.050	1.150	0.041	0.045
b	0.300	0.500	0.012	0.020
c	0.100	0.200	0.004	0.008
D	2.820	3.020	0.111	0.119
E	1.500	1.700	0.059	0.067
E1	2.650	2.950	0.104	0.116
e	0.950(BSC)		0.037(BSC)	
e1	1.800	2.000	0.071	0.079
L	0.300	0.600	0.012	0.024
θ	0°	8°	0°	8°

- SC-70-5 PACKAGE OUTLINE DIMENSIONS



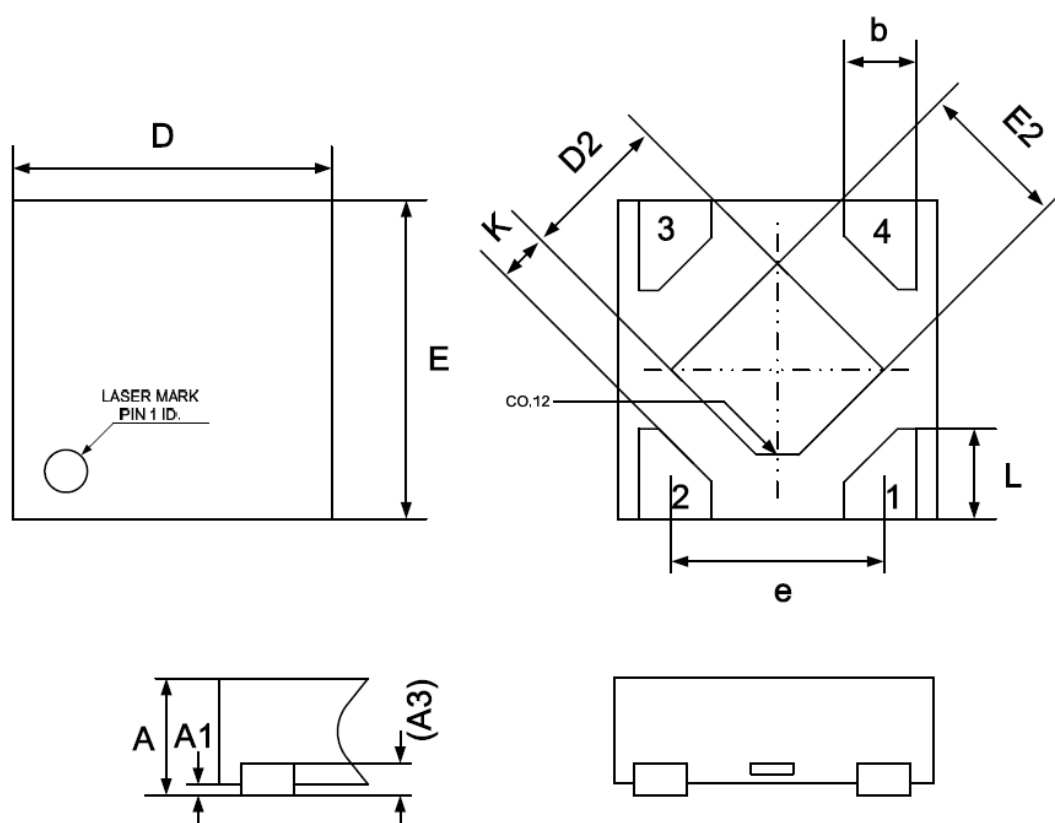
Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	0.800	1.100	0.031	0.044
A1	0.000	0.100	0.000	0.004
B	1.150	1.350	0.045	0.054
b	0.150	0.400	0.006	0.016
C	1.800	2.450	0.071	0.096
D	1.800	2.250	0.071	0.089
e	0.650		0.026	
H	0.080	0.260	0.003	0.010
L	0.210	0.460	0.008	0.018

- SOT-89-3 PACKAGE OUTLINE DIMENSIONS



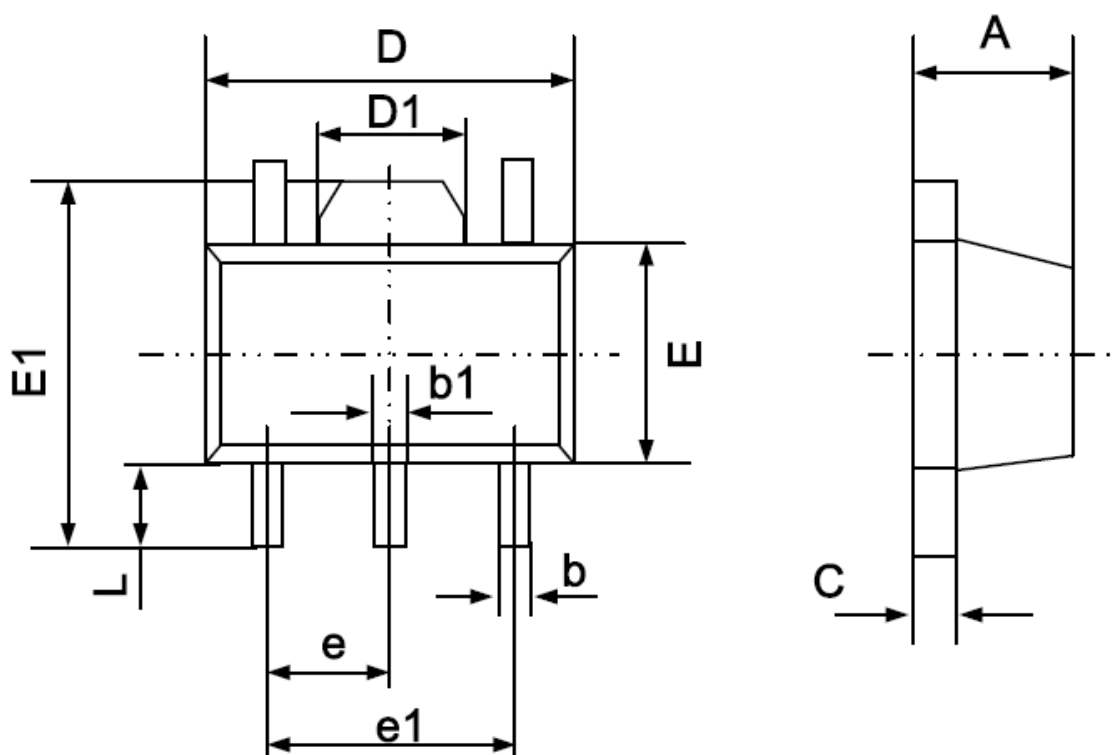
Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	1.400	1.600	0.055	0.063
b	0.320	0.520	0.013	0.020
b1	0.400	0.580	0.016	0.023
c	0.350	0.440	0.014	0.017
D	4.400	4.600	0.173	0.181
D1	1.550 REF		0.061 REF	
E	2.300	2.600	0.091	0.102
E1	3.940	4.250	0.155	0.167
e	1.500 TYP		0.060 TYP	
e1	3.000 TYP		0.118 TYP	
L	0.900	1.200	0.035	0.047

● DFN1X1-4 PACKAGE OUTLINE DIMENSIONS



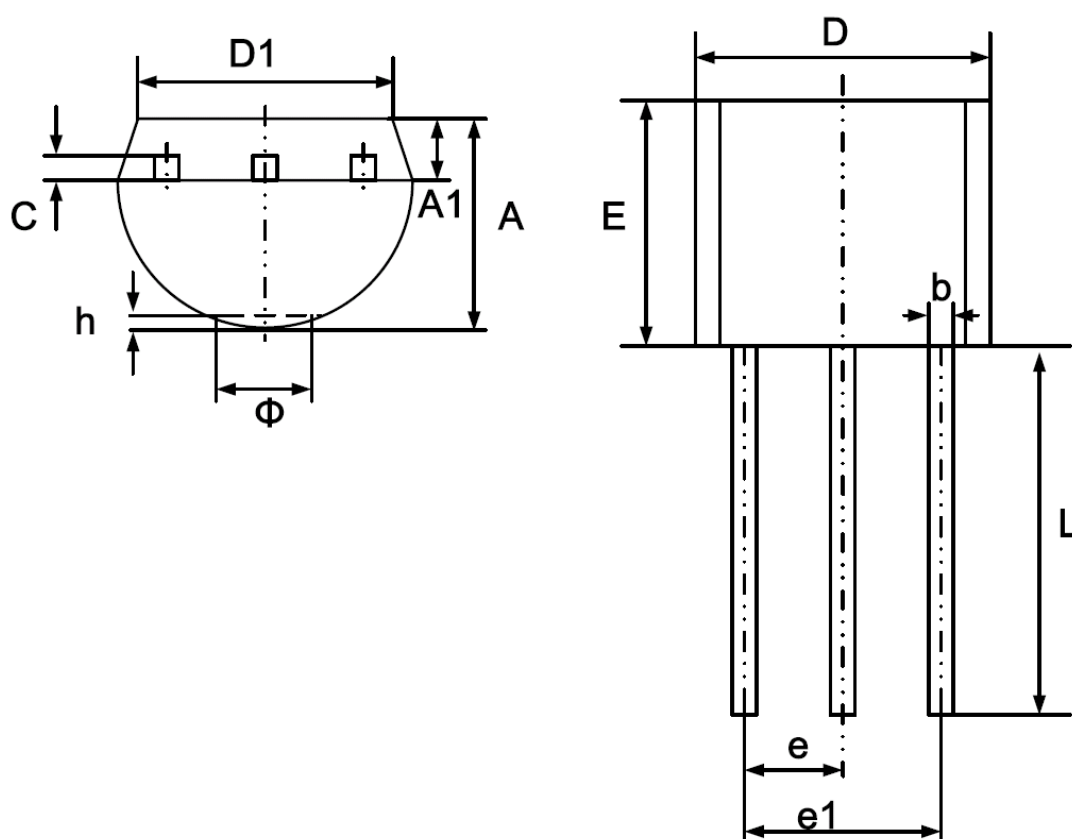
Symbol	Dimensions In Millimeters		
	Min.	Nom.	Max.
A	0.340	0.370	0.400
A1	0.000	0.020	0.050
A3	0.100REF		
b	0.170	0.220	0.270
D	0.950	1.000	1.050
E	0.950	1.000	1.050
D2	0.430	0.480	0.530
E2	0.430	0.480	0.530
L	0.200	0.250	0.300
e	-	0.650	-
K	0.150	-	-

- SOT-89-5 PACKAGE OUTLINE DIMENSIONS



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	1.400	1.600	0.055	0.063
b	0.320	0.520	0.013	0.020
b1	0.360	0.560	0.014	0.022
c	0.350	0.440	0.014	0.017
D	4.400	4.600	0.173	0.181
D1	1.400	1.800	0.055	0.071
E	2.300	2.600	0.091	0.102
E1	3.940	4.250	0.155	0.167
e	1.500 TYP		0.060 TYP	
e1	2.900	3.100	0.114	0.122
L	0.900	1.100	0.035	0.043

- TO-92 PACKAGE OUTLINE DIMENSIONS



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	3.300	3.700	0.130	0.146
A1	1.100	1.400	0.043	0.055
b	0.380	0.550	0.015	0.022
c	0.360	0.510	0.014	0.020
D	4.400	4.700	0.173	0.185
D1	3.430		0.135	
E	4.300	4.700	0.169	0.185
e	1.270 TYP		0.050 TYP	
e1	2.440	2.640	0.096	0.104
L	14.100	14.500	0.555	0.571
Φ		1.600		0.063
h	0.000	0.380	0.000	0.015

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